



DEPARTMENT OF CIVIL ENGINEERING
Indian Institute of Technology Guwahati

Subject: Civil Engineering Materials (CE 211)

Date: 22/11/2021

Time: 9:00 - 12:00 Hrs

Answer All Questions (Briefly to the point)

Total Marks: 75

- 1a) What are the advantages of using air entraining admixtures in concrete? [3]
- b) How does the maximum size of aggregate (MSA) affect the workability of concrete mix at a given water content? [3]
- c) Explain about the effect of w/c ratio on permeability of concrete. [3]
- d) From viewpoint of compressive strength of concrete, explain with reasons, which of the following two options is undesirable: [3]
- i) Concrete cast at 5 °C and cured at 21 °C.
- ii) Concrete cast at 21 °C and cured at 5 °C.
- e) Explain about different forms of chloride in concrete and their effect on corrosion of steel reinforcement in concrete. [4]
- f) What is meant by polarization in the context reinforcing steel corrosion in concrete? How is it related to corrosion rate of reinforcing steel in concrete? [3]
- g) What are the functions of different electrodes used during linear polarization resistance (LPR) measurement with guard ring arrangement to determine the corrosion current density of steel reinforcement in concrete? [6]
- h) Explain about the difference between cement-lime mortar, and mortar cement mortar for masonry. [3]
- 2) From the following information, design a concrete mix by Indian Standard guidelines (IS 10262 : 2019):
- Reinforced concrete subjected to 'moderate' exposure condition.
 - Grade of concrete: M 25.
 - Slump of concrete = 120 mm.
 - Cement: ordinary Portland cement (OPC) 53 grade, Specific gravity of cement = 3.15.
 - Nominal maximum size of aggregate = 20 mm.
 - Type of coarse aggregate: sub-angular aggregate, Fine aggregate (sand): confirming to Grading Zone II.
 - Specific gravity (SSD) of coarse aggregate = 2.69, Specific gravity (SSD) of fine aggregate = 2.65.
 - Reduction in water content due to use of superplasticizer = 27%.
 - Dosage of superplasticizer = 0.9% by mass of cement content, Specific gravity of superplasticizer = 1.21.
- i) Calculate the quantities of water, cement, superplasticizer, sand, and coarse aggregate for 1 m³ of the concrete mix. [12]
- ii) Also, determine the revised quantities of the ingredients for 1 m³ of the concrete mix, if water absorption of coarse aggregate, and surface moisture of sand are 0.5%, and 1.4% respectively. [3]
- 3) Linear polarization resistance (LPR) test was carried out on a chloride contaminated reinforced concrete specimen to determine the corrosion current density of steel reinforcement. The reinforced concrete specimen is embedded with a steel bar of diameter 12 mm. The applied perturbation is confined to top half surface area of the steel bar for a length of 30 cm. The polarization resistance of the steel bar obtained from LPR test is 125 Ohm. Calculate the corrosion current density of the embedded steel bar in $\mu\text{A}/\text{cm}^2$. [4]

End-Semester Examination (July - November 2021)

- 4a)** Why is blend of ordinary Portland cement (OPC) and fly ash suitable for minimizing the sulfate attack in concrete? **[4]**
- b)** Why is curing of concrete important? **[3]**
- c)** Explain about the mechanism of carbonation induced corrosion of steel reinforcement in concrete. **[3]**
- d)** How does permeability of concrete affect its durability? **[3]**
- e)** Explain about the effect of cement type on chloride induced corrosion of steel reinforcement in concrete. **[4]**
- f)** Describe briefly about gravimetric (mass loss) measurement for determining the corrosion rate of steel reinforcement in concrete. **[4]**
- g)** What are the functions of different components of cement-lime mortar used for masonry? **[4]**
- h)** What is the significance of fiber saturation point of wood? **[3]**

IS 10262 : 2019

Table 1 Value of X
(Clause 4.2)

Sl No.	Grade of Concrete	Value of X
(1)	(2)	(3)
i)	M10 } M15 }	5.0
ii)	M20 } M25 }	5.5
iii)	M30 } M35 } M40 } M45 } M50 } M55 } M60 }	6.5
iv)	M65 and above	8.0

IS 10262 : 2019

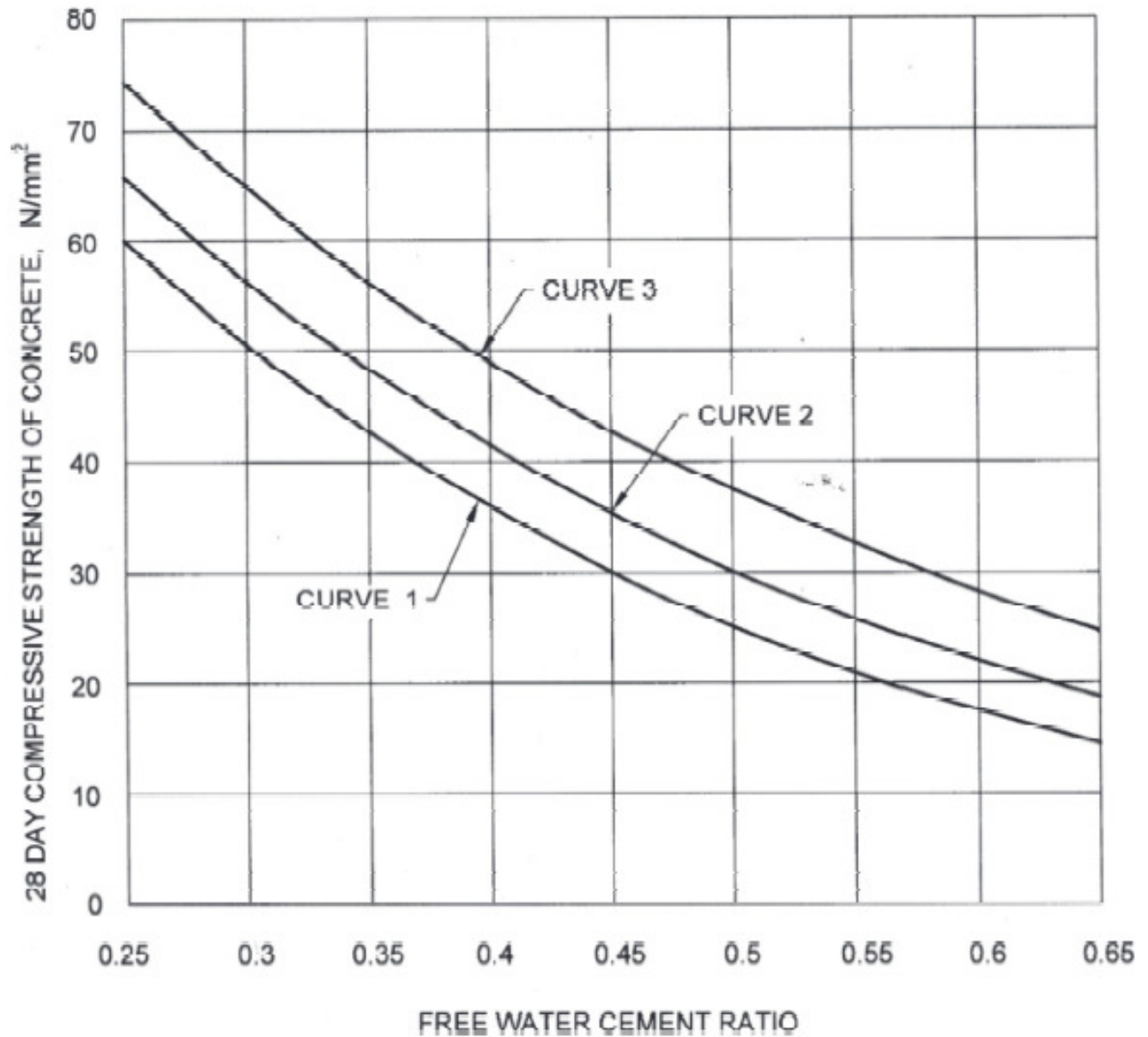
Table 2 Assumed Standard Deviation
(Clause 4.2.1.3)

Sl No.	Grade of Concrete	Assumed Standard Deviation N/mm ²
(1)	(2)	(3)
i)	M10 } M15 }	3.5
ii)	M20 } M25 }	4.0
iii)	M30 } M35 } M40 } M45 } M50 } M55 } M60 }	5.0
iv)	M65 } M70 } M75 } M80 }	6.0

NOTES

1 The above values correspond to good degree of site control having proper storage of cement; weigh batching of all materials; controlled addition of water; regular checking of all materials; aggregate grading and moisture content; and regular checking of workability and strength. Where there are deviations from the above, the site control shall be designated as fair and the values given in the above table shall be increased by 1 N/mm².

2 For grades M65 and above, the standard deviation may also be established by actual trials based on assumed proportions, before finalizing the mix.



Curve 1 : for expected 28 days compressive strength of 33 and $< 43 \text{ N/mm}^2$.
Curve 2 : for expected 28 days compressive strength of 43 and $< 53 \text{ N/mm}^2$.
Curve 3 : for expected 28 days compressive strength of 53 N/mm^2 and above.

NOTES

1 In the absence of data on actual 28 days compressive strength of cement, the curves 1, 2 and 3 may be used for OPC 33, OPC 43 and OPC 53, respectively.

2 While using PPC/PSC, the appropriate curve as per the actual strength may be utilized. In the absence of the actual 28 days compressive strength data, curve 2 may be utilized.

FIG 1. RELATIONSHIP BETWEEN FREE WATER CEMENT RATIO AND 28 DAYS COMPRESSIVE STRENGTHS OF CONCRETE FOR CEMENTS OF VARIOUS EXPECTED 28 DAYS COMPRESSIVE STRENGTHS

IS 456 : 2000 (Reaffirmed 2021)

Table 5 Minimum Cement Content, Maximum Water-Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregates of 20 mm Nominal Maximum Size

(Clauses 6.1.2, 8.2.4.1 and 9.1.2)

Sl No.	Exposure	Plain Concrete			Reinforced Concrete		
		Minimum Cement Content kg/m ³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete	Minimum Cement Content kg/m ³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete
1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Mild	220	0.60	—	300	0.55	M 20
iii)	Moderate	240	0.60	M 15	300	0.50	M 25
iii)	Severe	250	0.50	M 20	320	0.45	M 30
iv)	Very severe	260	0.45	M 20	340	0.45	M 35
v)	Extreme	280	0.40	M 25	360	0.40	M 40

NOTES

1 Cement content prescribed in this table is irrespective of the grades of cement and it is inclusive of additions mentioned in 5.2. The additions such as fly ash or ground granulated blast furnace slag may be taken into account in the concrete composition with respect to the cement content and water-cement ratio if the suitability is established and as long as the maximum amounts taken into account do not exceed the limit of pozzolona and slag specified in IS 1489 (Part 1) and IS 455 respectively.

2 Minimum grade for plain concrete under mild exposure condition is not specified.

IS 10262 : 2019

Table 3 Approximate Air Content

(Clause 5.2)

Sl No.	Nominal Maximum Size of Aggregate mm	Entrapped Air, as Percentage of Volume of Concrete
(1)	(2)	(3)
i)	10	1.5
ii)	20	1.0
iii)	40	0.8

IS 10262 : 2019

**Table 4 Water Content per Cubic Metre of
Concrete For Nominal Maximum Size of
Aggregate
(Clause 5.3)**

Sl No.	Nominal Maximum Size of Aggregate mm	Water Content ¹⁾ kg
(1)	(2)	(3)
i)	10	208
ii)	20	186
iii)	40	165

¹⁾Water content corresponding to saturated surface dry aggregate.

NOTES

1 These quantities of mixing water are for use in computing cement/cementitious materials content for trial batches.

2 On account of long distances over which concrete needs to be carried from batching plant/RMC plant, the concrete mix is generally designed for a higher slump initially than the slump required at the time of placing. The initial slump value shall depend on the distance of transport and loss of slump with time.

IS 10262 : 2019

**Table 5 Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine
Aggregate for Water-Cement/Water-Cementitious Materials Ratio of 0.50
(Clause 5.5)**

Sl No.	Nominal Maximum Size of Aggregate mm	Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate			
		Zone IV	Zone III	Zone II	Zone I
(1)	(2)	(3)	(4)	(5)	(6)
i)	10	0.54	0.52	0.50	0.48
ii)	20	0.66	0.64	0.62	0.60
iii)	40	0.73	0.72	0.71	0.69

NOTES

1 Volumes are based on aggregates in saturated surface dry condition.

2 These volumes are for crushed (angular) aggregate and suitable adjustments may be made for other shape of aggregate.

3 Suitable adjustments may also be made for fine aggregate from other than natural sources, normally, crushed sand or mixed sand may need lesser fine aggregate content. In that case, the coarse aggregate volume shall be suitably increased.

4 It is recommended that fine aggregate conforming to Grading Zone IV, as per IS 383 shall not be used in reinforced concrete unless tests have been made to ascertain the suitability of proposed mix proportions.